Voronoi-based Indoor Room Search and Rescue with Environmental Exploration

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***Abstract*—In this project, we investigate of autonomous robots for mapping, navigation, with the shortest path planning to the target. We implemented voronoi diagram for path planning and avoiding obstacles, we generated local points (way-points) and applied Travel Salesman Problem (TSP) Algorithms to connect the waypints by searching the closest edge of the voronoi diagram, we then applied the Dynamic Window Approach method (DWA) to allow the robot to navigate locally based on the TSP and Voronoi diagram paths. We formulated TSP algorithms by an integer linear programming and solved it by the Miller–Tucker–Zemlin (MTZ) formulation [2]. For time optimization, we used Goruobi modules as a solver for TSP[].**

***Index Terms*—Voronoi Diagram, Artificial Intelligent, local Navigation, Dynamic Window Approach method (DWA), Travel Salesman Problem.**

I. Project Outline

ENVIRONMENTAL Exploration: Construction of TWO kinds of Voronoi Diagram maps: Voronoi Diagram, and Generalized Voronoi Diagram (GVD) as follows. We do not need to implement and program VD, and GVD. There are some MATALB functions/commands for VD available. GVD might be available in MATALB as well, Or we may find online with MTALB file exchange, github, Sci-hub (https://en.wikipedia.org/wiki/Sci-Hub), or other resources.

5. Multi-goal path planning: Once the construction of Voronoi Diagram (GVD) maps is accomplished, some algorithms for TSP (Travelling salesman problem) is applied to connect some goals/waypoints that we need to explore, rescue, and reach. Running your TSP algorithm to connect some points such as green stars:

TSP can connect waypoints (injured persons, or some documents – we have known their GPS coordinates) in certain sequence to minimize the total traveling distance. Once sequence has been determined, robot needs to carry out path planning (PP) (navigation). • Robot will perform navigation/path planning. • Build up map (Mapping).

7. Local navigator using DWA (Dynamic Window Approach) method to walk a mobile robot from starting point to goal along the edges of built GVD. Tingjun Lei can help this task significantly to using DWA to navigate the robot to connect multiple goals depicted by the green stars.

II. Salesman Traveling Problem

The Traveling Salesman Problem (TSP) is a combinatorial optimization problem where the aim is to find the lowest cost route though a given set of nodes that returns to the initial node, forming a *cycle* or *tour* (a path that visits all the points or vertices, without repeating any vertices, before returning to its origin). It is named after the canvassing salesperson, whose goal is to visit a certain set of locations efficiently, before returning to their own home base. The TSP is a fundamental theoretical problem, and as it is NP-hard, it is related to one of the Clay Institute's Millennium Prize Problems [Coo11], [Coo06]. While a central academic problem, the TSP has a variety of uses outside of academia. Companies such as FedEx and UPS face the TSP daily, as they plan effective routes to deliver packages [Coo11].

We begin by formally defining the TSP. Given a graph $G = (V, E)$, where for each edge ij 2 E , let the cost cij be known. For any Hamiltonian cycle H of G, let $C(H) = P$ ij2H cij be the total cost. Then the Traveling Salesman Problem is to find the Hamiltonian cycle – a cycle passing through each vertex exactly once – such that C(H) is minimized. We provide details on different formulations in a later section. For details on the TSP, we refer $\forall x$

III. one minute paper

One minute summary: `the MATLAB code` the code is

A. 1st Stage

is the VD (the black line in the fig.1).

B. 2nd Stage

generate manually local points (waypoints), the green points, by adding its coordinates to the map to simulate the TSP/MTZ.

C. 3rd Stage

run the TSP, then using the sequence generated from the TSP output it will connect the waypoints, the green points, which is the red line in the figure. We try to search the closest edge of the Voronoi diagram.

D. 4th Stage

DWA method will breakdown the path, the red line specifically between waypoints, to multiple points that DWA can navigate locally.

IV. Objectives

V. Project Memo

1) Environmental Exploration: Construction of TWO kinds of Voronoi Diagram maps: Voronoi Diagram, and Generalized Voronoi Diagram (GVD) as follows. We do not need to implement and program VD, and GVD.

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2) Multi-goal path planning: Once the construction of Voronoi Diagram (GVD) maps is accomplished, some algorithms for TSP (Travelling salesman problem) is applied to connect some goals/waypoints that we need to explore, rescue, and reach. Running your TSP algorithm to connect some points such as green stars:

3) TSP (Travelling salesman problem): you may study TSP here

4) Local navigator using DWA (Dynamic Window Approach): method to walk a mobile robot from starting point to goal along the edges of built GVD.

5) 2D Local Grid Map: I wish you both can then build up a 2D map by following "2D Local Grid Map" https://www.mathworks.com/matlabcentral/fileexchange/52749-2d-local-grid-map

VI. Project Phase Stages

1) Environmental Exploration:
2) Multi-goal path planning:
3) Local navigator:
4) Mapping:

VII. Psudo Code

Algorithm 1 Calculate $y = x^n$

Require: $n \geq 0 \vee x \neq 0$
Ensure: $y = x^n$
  $y \leftarrow 1$
  if $n < 0$ **then**
    $X \leftarrow 1/x$
    $N \leftarrow -n$
  else
    $X \leftarrow x$
    $N \leftarrow n$
  end if
  while $N \neq 0$ **do**
    if N is even **then**
      $X \leftarrow X \times X$
      $N \leftarrow N/2$
    else {N is odd}
      $y \leftarrow y \times X$
      $N \leftarrow N - 1$
    end if
  end while

VIII. Conclusion

The conclusion goes here.

Appendix A
Proof of the First Zonklar Equation

Appendix one text goes here.

Appendix B

Appendix two text goes here.

Acknowledgment

The authors would like to thank...

References

[1] H. Kopka and P. W. Daly, *A Guide to LaTeX*, 3rd ed. Harlow, England: Addison-Wesley, 1999.
[2] 0.1145/321043.321046 Miller, C. E. and Tucker, A. W. and Zemlin, R. A.

PLACE PHOTO HERE

Michael Shell Biography text here.

John Doe Biography text here.

Jane Doe Biography text here.